

Global sensitivity analysis of correlated uncertainties in life cycle assessment

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[Correction added on October 8, 2025 after first online publication: The copyright line was changed.]

Abstract

Recent advances in research have made global sensitivity analysis of very large and highly linear life cycle assessment systems feasible. In this paper, we build on these developments to include sensitivity analysis of correlated parameters and nonlinear models. We augment numerical uncertainty propagation with Monte Carlo simulations (i) to include propagation of uncertainty from uncertain variables in parameterized inventory datasets; (ii) to account for correlations between process inputs and outputs and in particular incorporate the carbon balance of combustion activities; (iii) to employ published time-series data instead of static values for electricity generation market mixes in Europe; (iv) to ensure that inputs which are supposed to reach a fixed total (e.g., the percentage contributions of power sources to an electricity mix) actually do so consistently by using the Dirichlet distribution. We then iterate on existing global sensitivity analysis protocols for high-dimensional systems to improve their computational performance. To correctly calculate sensitivity rankings for correlated inputs, we use SHapley Additive exPlanations as feature importance metrics with gradient boosted trees. Our results for a case study of climate change impacts of an average Swiss household confirm that neglecting correlations limits the validity of uncertainty and sensitivity analysis. Our methodology and correlated sampling modules are given as open source code.

KEYWORDS

correlated sampling, ENTSO-E time-series electricity data, global sensitivity analysis, industrial ecology, parameterized life cycle inventory, Shapley values

1 | INTRODUCTION

Monte Carlo (MC) simulations are widely used in the uncertainty analysis of life cycle assessment (LCA) models. When integrated into global sensitivity analysis (GSA) methods, they help identify key drivers of uncertainty in environmental impact estimates by assessing how simultaneous variations in multiple inputs affect model outputs (Hauschild et al., 2018). Unlike local sensitivity methods, which evaluate the influence of one input at a time, global methods provide a more comprehensive understanding of the system. These analyses enhance our understanding of LCA robustness—the ability of LCA models to yield consistent results despite various assumptions and parameter variations. While most MC simulations are based on independent sampling, it has been long recognized that correlated inputs and the underlying system logic should be taken into

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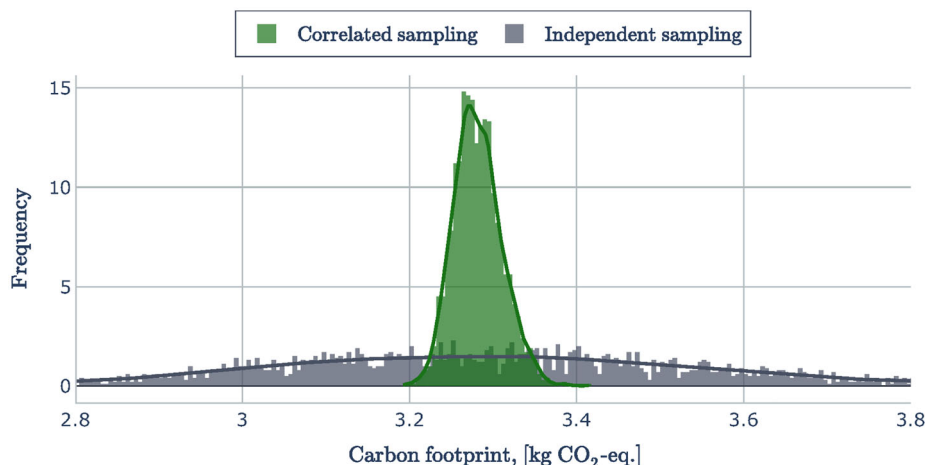


FIGURE 1 Uncertainty distributions of the carbon footprint for 1 kg of basmati rice from 2000 MC simulations. Results compare independent and correlated sampling, where the shares follow lognormal distributions: $S_{IN} \sim \text{Lognormal}(\log(0.7), 0.1)$ and $S_{RoW} \sim \text{Lognormal}(\log(0.3), 0.1)$. Underlying data for this figure are available in Supporting Information S3.

account for thorough uncertainty and sensitivity analysis (Groen & Heijungs, 2017; Smith et al., 1992). This is important because in real-world systems, variables often exhibit correlations, meaning that changes in one process can influence and correlate with others. By including correlations, the analysis becomes more reflective of the true behavior of the system, allowing for more reliable decision-making in the presence of uncertainties.

To illustrate the effect of independent and correlated sampling, consider this simplified example: The goal is to calculate the carbon footprint of 1 kg of basmati rice. The global market for basmati rice is composed of a median share of 70% from India, with an emission coefficient of 3.4 kg CO₂-eq per kg, and a median share of 30% from the Rest of the World (RoW), with an emission coefficient of 3.0 kg CO₂-eq per kg. The respective shares S_{IN} and S_{RoW} vary between years: They depend on annual production volumes in these regions, which vary from year to year and are thus uncertain. In the ecoinvent database, these shares would typically be modeled independently as lognormal distributions. Therefore, the total carbon footprint is calculated as $3.4\text{kg} \cdot S_{IN} + 3.0\text{kg} \cdot S_{RoW}$. However, this may lead to unreasonable results, as the sum of both shares may not add up to 100%. By contrast, if we account for the relation between these shares, where $S_{IN} = 1 - S_{RoW}$, the total impact can instead be computed as $3.4 \cdot S_{IN} + 3.0 \cdot (1 - S_{IN})$. Figure 1 presents the uncertainty distributions of carbon footprints based on 2000 Monte Carlo simulations. It is evident that independent sampling produces a significantly wider distribution of the carbon footprint. Therefore, the uncertainty would be overestimated and may lead to misleading results. For example, when basmati rice were compared to other rice variations, a potentially significant difference may not be recognized as such, compromising the decision support that is delivered.

Despite their importance, correlations in LCAs are typically considered only in comparative settings, meaning that uncertainties are correlated when varying parameters common to the studied alternative products (Gregory et al., 2016). For instance, Geisler et al. (2005) used correlated Latin hypercube sampling to propagate uncertainty and variability in life cycle inventory (LCI) flows and characterization factors into impact scores for two plant protection products. The study found that, when uncertainties were properly accounted for, most impact category differences between the products were not significant. The authors highlight the need for routine quantitative uncertainty analyses in LCA and recommend that data providers include correlation estimates for individual parameters to improve assessment accuracy. Similarly, Henriksson et al. (2015) applied correlated sampling to compare the carbon footprint of large- and small-scale catfish farms in Vietnam. The study underscores that impact assessments are shaped by modeling choices and opportunistically collected data rather than rigorous statistical design, but dependent sampling helps mitigate biases by maintaining consistency in the underlying assumptions of comparative LCAs. Qin and Suh (2018) investigated the impact of sampling strategies on LCAs by comparing two randomly selected processes and two electricity-producing processes that share many common datasets. The study highlights that errors from independent sampling are more pronounced when products have numerous shared processes, increasing the likelihood of erroneous conclusions if correlations are ignored. While the aforementioned studies conclude that proper treatment of correlated uncertainties establishes a fair basis for product comparison, they do not address correlations and dependencies present within individual product systems in non-comparative LCAs.

Including correlations is in general difficult and, perhaps, even impossible if suitable empirical data are not available. Due to limited data availability, challenges in prioritizing data collection, and lack of appropriate tools, LCA databases typically only provide uncertainty estimates for individual datasets, but fail to reveal how uncertainties across different datasets are connected or influence each other. In technical terms, this means that dataset uncertainties are modeled independently, ignoring their interdependence; that is, joint distributions between datasets are generally excluded. As the demand for more accurate measurement of resource consumption, energy usage, and waste generation grows in a data-driven context, increasingly more sources of data emerge that often times reveal implicit correlations. For instance, the European Network of Transmission System Operators for Electricity (ENTSO-E) provides fundamental information about load, generation, transmission, and balancing of electricity (Hirth et al., 2018). The time series of electricity generation data from different technologies can be used in LCI as MC samples for regional electricity

mixes. Naturally, it contains correlations; for example, the share of imported electricity is negatively correlated with the amount of electricity produced from intermittent renewable sources. Despite its availability and abundance, real measurements have not been systematically mapped to electricity production life cycle datasets yet.

Correlations are also present in inventories comprising a mix of activities with a fixed total. In some fields, data involving proportions, percentages, or parts of a whole are known as compositional data. One example is relative electricity grid mixes, where all power contributions add up to 100%. Another example is the import of natural gas, which includes offshore and onshore shares of petroleum and gas production that must also add up to 100%. In such cases, database managers typically remove already defined uncertainty distributions to ensure that independent MC sampling does not produce totals below or above the reference total. For instance, the commonly used LCI database ecoinvent removes uncertainties from such datasets labeled as markets (Weidema et al., 2013; Wernet et al., 2016). By choosing a multivariate distribution instead, it would be possible to ensure fixed totals while preserving uncertainties in individual exchanges. Yet, typical uncertainty analysis in LCA does not account for relationships or interactions between multiple variables simultaneously, in other words does not utilize multivariate distributions.

Although accounting for correlations is inherently challenging and feasible only in selected processes as a result of limited data availability and methodological constraints, enhancing system logic with *causal* dependencies has been widely addressed through parameterized inventories (Cooper et al., 2012). The ecoinvent database alone contains hundreds of parameterized datasets with readily available formulas to calculate inventory flow amounts from sets of parameters (Weidema et al., 2013; Wernet et al., 2016). Often times, these parameters have a predefined uncertainty distribution that can be propagated to inventory flows and impact assessment results. Also, LCA researchers create increasingly more parameterized LCI models with some of the examples listed below:

- Niero et al. (2014) parameterized wooden pallets as a function of their configuration, mass, and manufacturing processes;
- Besseau et al. (2019); Sacchi et al. (2019) provided geographical, spatial, and technological parameterization of the wind energy sector in Denmark;
- Köhler et al. (2007) modeled industrial wastewater treatment systems based on wastewater composition and process parameters;
- Sacchi et al. (2022) represented passenger vehicles depending on various powertrain categories, vehicle sizes, and fuel types.

Furthermore, there exist causal relationships in non-parameterized inventories, where mass, monetary, and physical balances should be preserved. For example, amounts of combusted fuel and emitted carbon dioxide in the same process are currently sampled independently from each other, despite the well-known stoichiometric reaction between fuel combustion and emissions. Despite the plethora of readily available dependencies, they are not systematically studied in uncertainty and sensitivity analysis of LCA.

An additional barrier is methodological. There are only a few GSA protocols, which can be used in LCA systems with hundreds of thousands of uncertain parameters (Cucurachi et al., 2021; Kim et al., 2022). The existing protocols have not been designed for correlated uncertainty propagation and sensitivity analysis. To include correlations and nonlinear dependencies, the underlying assumptions of GSA protocols should be revised.

In this work, we aim to close research gaps associated with GSA of LCA in the presence of causal nonlinear dependencies and correlations on a case study of carbon footprint from Swiss consumption. We outline the main contributions of this work below.

- (i) We demonstrate the process of correlated and dependent uncertainty propagation using four distinct sampling modules, each with its unique underlying sampling mechanism. Namely, we include ecoinvent datasets parameterized with variables and formulas; carbon balancing in combustion of liquid fuels; ENTSO-E electricity measurements used as LCI; and uncertainty in ecoinvent markets modeled with the multimodal Dirichlet distribution that ensures the “fixed total” constraint.
- (ii) We construct multi-step GSA protocol to confidently tackle high-dimensional LCA models with correlated inputs by combining the computational merits of Cucurachi et al. (2021), GSA performance of Kim et al. (2021), and extreme gradient boosting (XGBoost) with SHapley Additive exPlanations (SHAP) values from the field of interpretable machine learning (ML) (Lundberg & Lee, 2017).
- (iii) We compare outcomes of MC simulations and GSA based on independent and correlated sampling.

All the computations relevant for this publication were performed with BRIGHTWAY version 2.5 software for advanced LCA computations that allows to seamlessly include new sampling modules. Our implementation can be found in the open source MIT-licensed Python repository AKULA (Kim, 2024c).

2 | METHODS

2.1 | GSA for matrix-based LCA models

GSA refers to a family of techniques that identify model inputs that are responsible for most of the uncertainty in the model output (Saltelli et al., 2008). They operate on models $y = g(x)$ that can be treated as a black box, with k inputs $x \in \mathbb{R}^k$ and p outputs $y \in \mathbb{R}^p$. Matrix-based LCA models

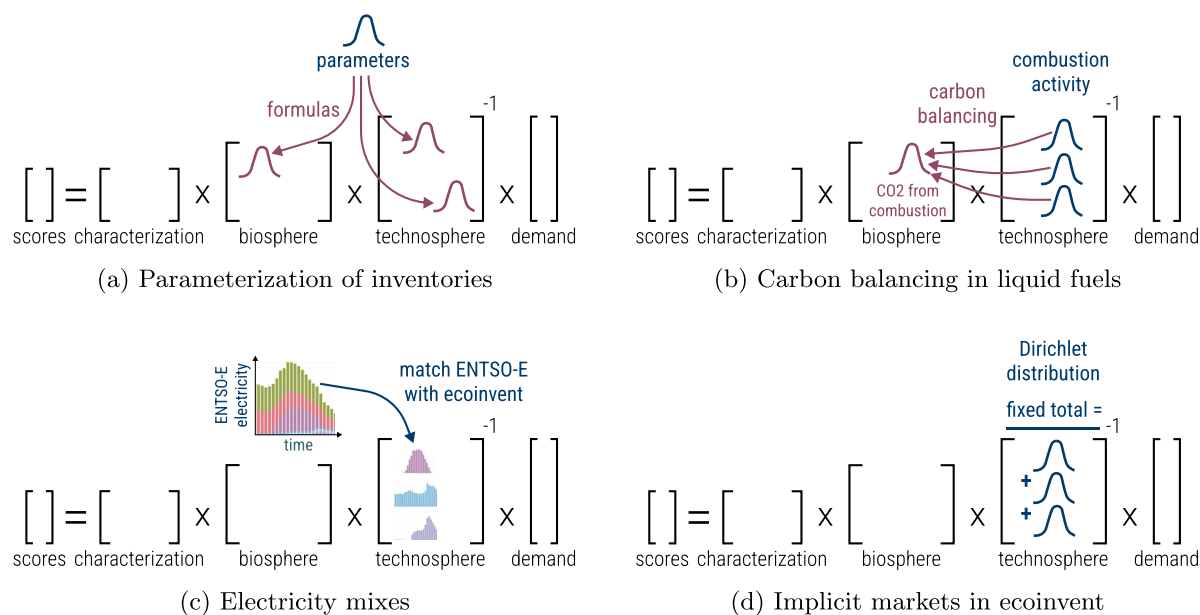


FIGURE 2 Schematic representation of sampling modules for the matrix-based life cycle assessment model. Independent elements are given in blue, and dependent elements in pink. ENTSO-E stands for the European Network of Transmission System Operators for Electricity.

$y = CBT^{-1}f$ can be expressed as multiplication of characterization $C \in \mathbb{R}^{p \times m}$, biosphere $B \in \mathbb{R}^{m \times n}$, and technosphere $T \in \mathbb{R}^{n \times n}$ matrices, scaled by the functional unit vector $f \in \mathbb{R}^n$ (Heijungs & Suh, 2002). Technosphere contains energy and material inventories for manufacturing of n products, biosphere is used to quantify m emissions to and from the natural environment, while characterization matrix collects these emissions into p life cycle impact assessment (LCIA) categories by assigning weights to relevant emissions. The product of interest is selected in the functional unit, and the output y gives the estimated environmental impacts, or LCIA scores. Uncertain characterization factors, biosphere and technosphere flows (or exchanges), and parameters in parameterized datasets act as inputs in a vector $x \in \mathbb{R}^k$. For the sake of notation simplicity, we address only one impact category (thereby, $p = 1$ and $y \in \mathbb{R}$).

Sensitivity indices are a quantitative measure of an input's importance. They are computed for each input either analytically for simpler models or numerically using MC simulations. Complete LCA models typically have hundreds of thousands of uncertain inputs. The combination of very high dimensionality and short execution times makes LCA models a challenging but feasible task for GSA. To date, only MC simulations coupled with numerical estimation methods can effectively address such problems. Numerical estimation uses computational algorithms to approximate sensitivity indices by simulating various inputs and analyzing their impact on the output. Unlike analytical approaches, which rely on mathematical equations to derive exact solutions, numerical methods can handle complex, non-linear systems without simplifying assumptions, though they tend to require more computational resources. Consequently, this is the approach we adopt in this study.

2.2 | Causation and correlations in MC simulations

In this section, we describe four modules that introduce causation and correlations into MC simulations. Each module is designed to address specific aspects of the sampling process. In this context, the modules add causation and correlations to the simulations, managing how variables interact and influence one another. Together, these modules enhance the accuracy and plausibility of the simulations. While they do not fully encompass the variety of possible interdependencies, in this study, they were chosen to demonstrate distinct sampling mechanisms that can be further extended to other datasets. Figure 2 provides their schematic representation.

2.2.1 | Parameterization of inventories

Parameterization refers to the practice of expressing inventory dataset relationships through formulas applied to named variables (Mueller et al., 2004). An example of a parameterized flow in a “passenger car” dataset of ecoinvent is brake wear emissions of the whole vehicle. The emissions are computed as the multiplication of two uncertain parameters: brake wear emissions per kilogram of vehicle and total vehicle weight. Then, heavier cars will tend to cause higher emissions and vice versa. Another example is in a “maize silage production” dataset, where the relation between water

content of maize and water in wet mass and dry mass is explicitly accounted for. Parameterization enhances data consistency and transparency by showing the origins of computed values (Cooper et al., 2012). It allows us to incorporate more complex and accurate models, as well as technological, geographical, and temporal variability. By propagating parameter variations, we can account for relations that would otherwise be treated as independent (Sacchi et al., 2019).

In this manuscript, we evaluated the variables and formulas given in ecoinvent v3.8, system model “cutoff by classification” (Wernet et al., 2016). We first carefully parsed the formulas with named variables and identified 565 activities with 857 parameters in 7509 biosphere and 640 technosphere exchanges. The complete list is given in Supporting Information S2. Then we generated independent samples for all named variables following their defined uncertainty distributions and propagated them to technosphere and biosphere flows (see Figure 2a). Most of the datasets are related to land use, transport, heat, and agricultural activities, though there were also few exchanges in production processes, for example, in the production of concrete. The use of named variables introduced new inputs to the GSA problem but also removed some exchanges that became dependent on the named variables.

2.2.2 | Carbon balancing in liquid fuels

Combustion of most fuels produces CO₂; in the standard LCA formulation, the amount of fuel combusted and the corresponding CO₂ produced are not linked, and each value is sampled independently. The vast majority of carbon in fuels is completely oxidized, and therefore, provided the carbon content of fuels is known—as it is in ecoinvent—the amount of CO₂ is a simple calculation. However, such stoichiometric balances are typically not incorporated into parameterized inventories in ecoinvent. Instead, certain datasets provide information on the chemical properties of substances, allowing us to derive these relationships on our own. Consequently, we treat this category as a separate sampling module distinct from the parameterized inventories.

To include this stoichiometry in MC simulations, we searched for ecoinvent activities whose input names contained “market for diesel,” “diesel,” “market for petrol,” and “petrol,” and which produced “carbon dioxide, fossil.” This search resulted in 812 activities with 407 biosphere exchanges and 1403 technosphere exchanges, encompassing animal and crop production, transport and construction processes, as well as manufacturing of various products, such as glass, pulp, cement, paperboard, motor vehicles, and others. The exchanges are listed in Supporting Information S2. For each activity, we treated the fuel input as an independent parameter following its given distribution, and the CO₂ emissions as dependent parameters (see Figure 2b). We assumed that carbon is fully oxidized. To compute LCIA scores, we used the updated CO₂ amounts that were summed up in case multiple fuel types caused the same CO₂ emissions.

2.2.3 | Electricity mixes

Ecoinvent models electricity mixes as annual averages without uncertainty. Electricity consumption in LCA remains the subject of vigorous debate, with disputes on the use of marginal versus average mixes, certified versus total generation, and which types of load can be shifted in time (Weber et al., 2010). As our case study focuses on consumer behavior, and many consumer electrical loads can be temporally shifted, at least in the short term, we illustrate our approach to adding uncertainty to LCI by substituting the annual average with time-series data gathered from the ENTSO-E Transparency Platform API (Hirth et al., 2018).

First, the actual generation data per each production type was collected from the ENTSO-E platform for 2019, 2020, and 2021, and the electricity generation categories were matched and disaggregated to ecoinvent activities (see Section 1 of Supporting Information S1). Disaggregation was necessary in cases where ENTSO-E only listed one generation type (e.g., “Hydro Water Reservoir” was allocated to both alpine and non-alpine reservoirs) and was done using annual production volumes from ecoinvent as allocation factors. We used ENTSO-E data to overwrite the low-, medium-, and high-voltage market mixes for 32 European countries, while taking into account ENTSO-E cross-border physical flow data to replace imports in the ecoinvent high-voltage market mixes (see Figure 2c). In total, we added the time-series data to 821 technosphere exchanges in 87 ecoinvent electricity markets that are listed in Supporting Information S2. By default, ecoinvent markets did not contain uncertainties.

Switzerland is modeled as a special case in ecoinvent, where its long-term import contracts from specific generation sources are included in the Swiss production mix and correspondingly subtracted from the production mixes of exporting countries (Treuer & Bauer, 2016). Our approach uses the ENTSO-E data for all countries, including Switzerland. However, there are many smaller generators in Switzerland that are exempt from the ENTSO-E real-time reporting requirement and do not appear in the time-series data. Therefore, following electricitymap.org (electricityMap, 2017), we used annual production totals from the Swiss Ministry for Energy (BFE, 2022) to create a residual mix of generation technologies, which is used to model the difference between time-series generation data for specific generators and the Swiss production total. This residual mix is more than 90% hydropower.

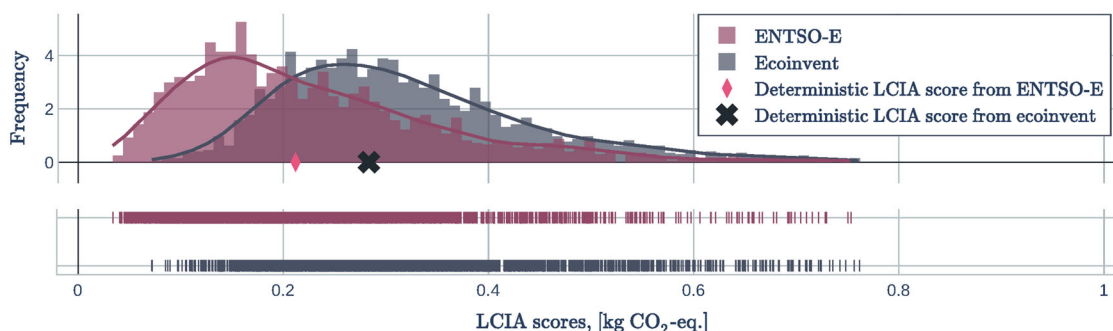


FIGURE 3 Histograms of climate change life cycle impact assessment (LCIA) scores from 1 kWh of low voltage electricity market in Denmark for two cases: (gray) uncertainties are only present in ecoinvent; (pink) ecoinvent uncertainties are complemented with the variations in electricity mixes derived from the Transparency Platform of the European Network of Transmission System Operators for Electricity (ENTSO-E). Underlying data for this figure are available in Supporting Information S3.

To illustrate the effect of adding ENTSO-E data, Figure 3 shows a histogram of climate change impacts of 1 kWh of low voltage electricity market in Denmark. In one case, uncertainties are only present in ecoinvent (gray), and in the other, they are complemented with variations in electricity mixes derived from ENTSO-E (pink). The differences in both deterministic and stochastic LCIA results are evident.

Naturally, the uncertainties in impacts would look differently if we select only day or nighttime ENTSO-E data, or if we focus on a particular season (winter, spring, summer, or autumn). One can also choose to follow the ecoinvent approach and add uncertainties directly to the annual averages in electricity mixes. Given the ENTSO-E data for the years 2019, 2020, and 2021, we can compute the averages for each year, fit a probability distribution to these three data points, and draw samples from the derived distributions. The choice of the most suitable approach depends on the goal and scope of each case study. In this work, we use the ENTSO-E time series without filtering for a particular time of the day or season, but we illustrate the effect of the other mentioned approaches in Section 2 of Supporting Information S1.

2.2.4 | Implicit markets in ecoinvent

In the ecoinvent database, the so-called market activities aggregate transforming activities that provide the same product in a given spatial and temporal context (Weidema et al., 2013; Wernet et al., 2016). This way one can conveniently draw directly from the market activity instead of calculating the respective shares manually. As market inputs should sum to a fixed total, and there is no way to specify input covariances or impose the “fixed total” constraint in datasets, uncertainty for market inputs is automatically removed by ecoinvent.

Markets are generated with an algorithm that looks for an exact match between the market product name and the reference product names of its inputs. However, there are cases where activities act like markets, but whose inputs do not have the exact same names. In this research article, we call them *implicit markets*. As these are not recognized as markets by the ecoinvent algorithm, their exchanges still contain uncertainty information. Therefore, during MC sampling, their totals incorrectly sum to a different number at each simulation. For example, the activity “natural gas, high pressure, import from DZ” in Switzerland provides one cubic meter of natural gas and has inputs of 0.69 cubic meters from “natural gas production” and 0.31 cubic meters of “evaporation of natural gas.” As both these inputs still have uncertainty, under standard MC sampling, their sum will deviate from precisely one cubic meter that this activity should in principle supply.

In this sampling module, we aim to identify the implicit markets with fuzzy string matching that looks for *inexact* matches between product names by utilizing Levenshtein distances (Levenshtein et al., 1966; Navarro, 2001). We then replace their uncertainty distributions with a multivariate Dirichlet distribution that can guarantee the “fixed total” constraint (see scheme in Figure 2d). While doing so, we ensure that marginal distributions in the market exchanges closely match the original uncertainty distributions defined by ecoinvent. We explain our approach in detail in Section 3 of Supporting Information S1. In total, we identified 61 implicit markets with 517 technosphere exchanges that we provide in Supporting Information S2.

Figure 4 shows an example of Dirichlet sampling in the implicit market “electricity for aluminium production.” As expected and by design, the mean values of Dirichlet samples are always well aligned with the predefined lognormal distribution means. The distribution widths, on the other hand, fit best to the input activities “heat and power co-generation from hard coal” and “heat and power co-generation from lignite” in Russia that have highest production volume shares for this market.

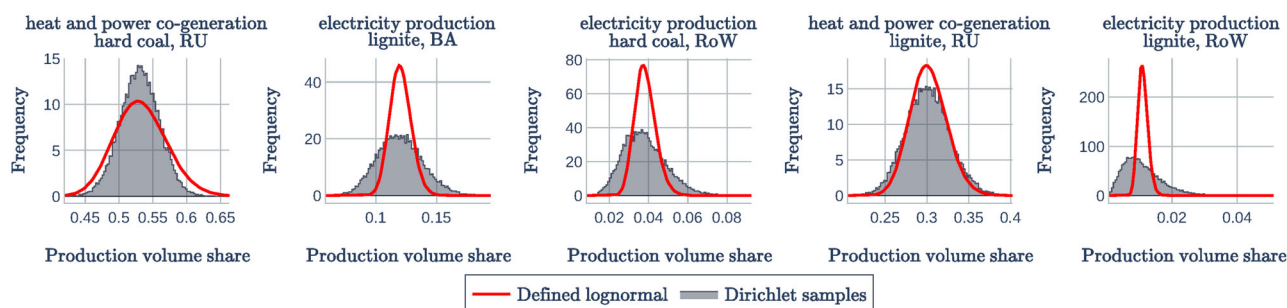


FIGURE 4 Uncertainties in the implicit market “high voltage electricity from coal for aluminium production” for European aluminium producing countries outside the region covered by the European Aluminium Association, that is, outside EU27 and EFTA. Dirichlet samples are shown in gray and ecoinvent lognormal distributions in red. The mean values of Dirichlet samples are always well aligned with the predefined (lognormal) distribution means. The distribution widths, on the other hand, fit best to the input activities that have highest production volume shares. Underlying data for this figure are available in Supporting Information S3. RU, Russia; BA, Bosnia and Herzegovina; RoW, Rest of the World.

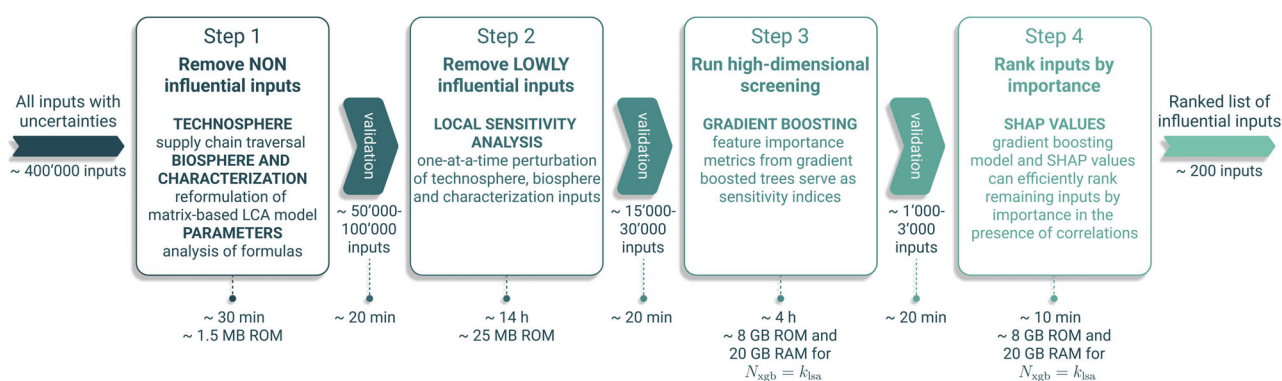


FIGURE 5 Flowchart of the proposed multi-step global sensitivity analysis (GSA) protocol for life cycle assessment (LCA) models. Number of inputs indicates how many uncertain inputs enter each GSA step. SHAP stands for the SHapley Additive exPlanations.

2.3 | Validation of Dirichlet distributions on electricity data

We validate the Dirichlet approach on the readily available ENTSO-E electricity data that was described in Section 2.2.3. We treat production volume shares as ground truth exchange amounts in all ENTSO-E time-series data for low-, medium-, and high-voltage electricity markets of each region. We then visually compare these distributions with samples from the Dirichlet approach and default ecoinvent approach that would typically use lognormal distributions. For the former, the distribution of the shares is used to derive means and variances for each electricity supplier in order to compute the parameters α and the scaling factor λ as described in Section 3 of Supporting Information S1. For the latter, we apply the maximum likelihood estimator to fit lognormal distribution to data and obtain its location, shape, and scale parameters. The results of this comparison for the Danish electricity markets are given in Section 3.1. Note that the sampling modules “electricity mixes” and “implicit markets” are generally independent of each other and are used together in this work solely for validation purposes.

2.4 | GSA protocol

When conducting GSA for complete LCA models, the total number of uncertain technosphere, biosphere, characterization, and parameter inputs can reach hundreds of thousands. To the extent of the authors’ knowledge, only the works of Cucurachi et al. (2021) and Kim et al. (2022) can tackle such high-dimensional LCA models. However, they are only suitable for independent inputs. In this section, we present an improved GSA protocol that can be employed in the presence of correlations and dependencies and can leverage the computational efficiency of Cucurachi et al. (2021), while retaining the main procedures and accuracy of Kim et al. (2022). Figure 5 gives an overview of the complete GSA protocol, where Steps 1–3 aim to gradually filter out non- and lowly influential inputs until only few thousand are left for ranking in Step 4. After each filtering step, we perform validation of sensitivity results as described in Section 2.5 to ensure that no important inputs are lost. The figure also shows approximate computation and memory demand of each procedure if the simulations are performed on a personal computer with the 11th Generation Intel Core

i5 processor and 32 GB of RAM. The implementation of this GSA protocol is provided in the Python repository AKULA (Kim, 2024c), with the main steps explained below.

Step 1: Remove non-influential inputs: We first try to understand whether the inputs have any effect on the output by analyzing the LCA model structure.

- For technosphere exchanges, we use supply chain traversal to compute contributions of technosphere flows to the total LCIA score (Cardellini et al., 2018) and remove exchanges with contributions less than a predefined threshold as was proposed by Cucurachi et al. (2021). This cutoff value is usually expressed as a percentage of the total impact score, where higher cutoffs disregard more inputs. In the original method, the cutoff was set to 0.1% of the total score (corresponds to the cutoff value of 0.001), which removes more than 99% of technosphere inputs and makes the subsequent steps significantly less computationally expensive. In this work, we employ the same approach, but set the cutoff to a rather conservative value of $1e^{-7}$ that filters out only 70%–80% of inputs. The motivation is to not dismiss potentially influential exchanges with low contributions but large uncertainty. Whether the cutoff is set correctly can be checked with the validation procedure described in Section 2.5.
- For biosphere and characterization inputs, we reformulate the LCA matrix-based model and remove inputs with zero contributions based on the structure of the technosphere, biosphere, and characterization matrices (Kim et al., 2022).
- Finally, we analyze formulas in the parameterized inventories and only keep parameters with non-zero contributions to the LCIA total score.

Step 2: Remove lowly influential inputs: Following the approach of Kim et al. (2022), we employ local sensitivity analysis to the inputs remaining after the Step 1 by perturbing them one-at-a-time and monitoring the effect of this perturbation on the LCIA score. Inputs with low effects will be discarded. Two perturbations per input are sufficient for linear input/output relations, as in biosphere and characterization case; the remaining inputs require at least three perturbations due to nonlinear effects. In this work, perturbations were done by multiplying and dividing the deterministic input value by a factor of 10. For each input, we compute variance of the LCIA score, and keep k_{lsa} inputs for the subsequent analysis. The parameter k_{lsa} should be chosen to be as low as possible to reduce computational burden of the subsequent steps but large enough to still completely capture variability in LCIA scores. Validation of sensitivity results described in Section 2.5 can help in tuning this parameter. After this step, the number of remaining inputs should be reduced to 10,000–30,000.

Step 3: Run high-dimensional screening: In line with Kim et al. (2021) and given that this LCA model is certainly nonlinear, in this step, we utilize gradient boosted trees and apply its XGBoost implementation to MC samples and LCIA scores (Chen & Guestrin, 2016; Friedman, 2001). When conducting MC simulations, we incorporate correlated and dependent sampling modules described in Section 2.2. The appropriate number of MC simulations N_{xgb} is in the range of k_{lsa} to $4k_{\text{lsa}}$. We perform grid search for parameter tuning in XGBoost, then train the model, and use feature importance metric as sensitivity index. The top k_{xgb} inputs with highest feature importance values are considered influential and will be ranked in the next step. Typical value for k_{xgb} after this step is in the order of 1000 to 3000 inputs.

Step 4: Rank inputs by importance: At this point, various sensitivity methods can be used since only few thousand inputs are left. In this work, we apply *Shapley values* that originate from coalitional game theory, and are based on the allocation of the total payout of a game to the contribution of each player. Here, payout refers to a model output, and players are model features. *Shapley effects* are sensitivity indices based on Shapley values, whose connection to the Sobol first and total order indices has been long recognized in GSA community (Owen, 2014). Since calculating Shapley effects is challenging due to large computational complexity, a number of works proposed techniques to efficiently estimate Shapley effects, while accounting for dependencies and interactions between inputs (Bénard et al., 2022; Iooss & Prieur, 2019; Plischke et al., 2021; Song et al., 2016). In this work, we employ a novel SHAP method and TreeSHAP algorithm that reduces the estimation of Shapley values from exponential to polynomial time if a tree-based prediction model is used (Lundberg & Lee, 2017; Lundberg et al., 2018). We combine SHAP with the XGBoost model trained on the previous step to efficiently rank inputs by importance. It has been argued that SHAP can indeed be applied in the presence of correlated inputs, making it a suitable method for our GSA protocol (Janzing et al., 2020).

2.5 | Validation of sensitivity results

An essential procedure throughout the presented GSA pipeline is to validate sensitivity results to ensure that no influential inputs are filtered out after any of the steps. The goal is to understand whether varying only a subset of inputs will result in the same variations of LCIA scores compared to varying all inputs. To this end, we perform 2000 MC simulations based on two sets of input samples: (i) unconditional variations of all inputs that result in LCIA scores Y_{all} , (ii) conditional variations of a subset of potentially influential inputs, such that they coincide with case (i), while non-influential inputs are set to their default values (Kim et al., 2021; Kim et al., 2022). The LCIA scores from case (ii) will be denoted as Y_{non} , Y_{lsa} , or Y_{xgb} based on which step in the GSA protocol they validate. We estimate the quality of sensitivity results by visually comparing scatter plots between the two sets of LCIA scores (Andres, 1997; Nossent et al., 2011; Tang et al., 2007). Narrower plots signify better agreement between LCIA scores, and hence, they point to correctly performed sensitivity analysis.

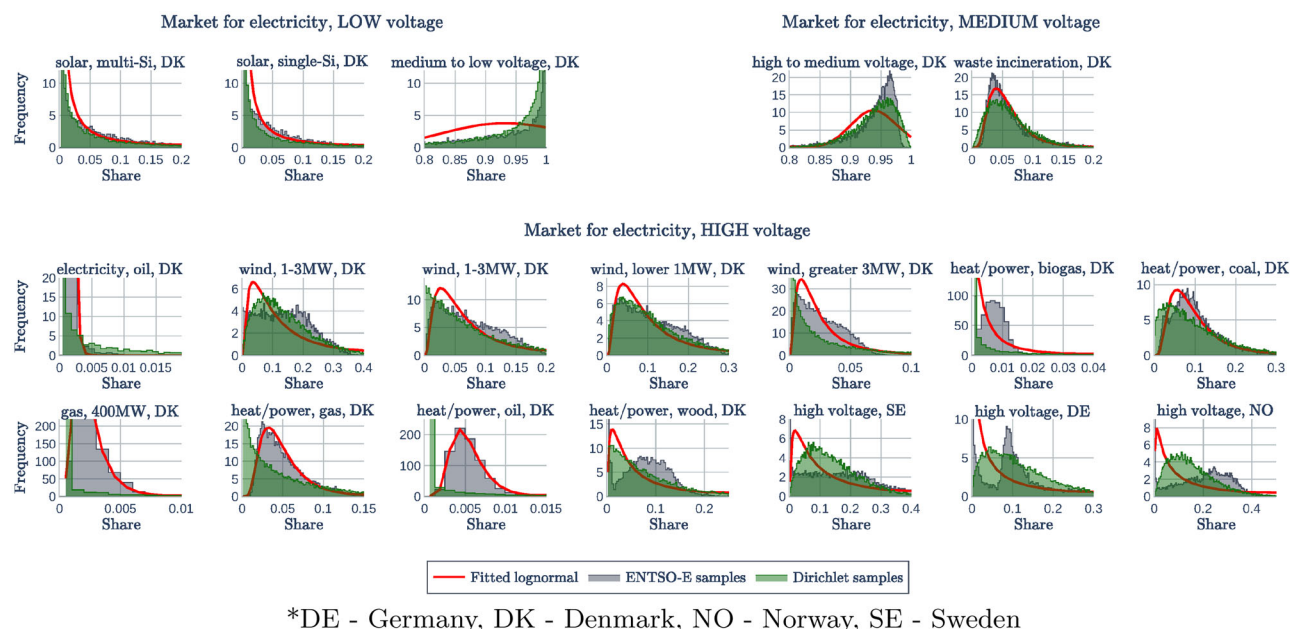


FIGURE 6 Validation of Dirichlet distribution on electricity mixes from the Transparency Platform of the European Network of Transmission System Operators for Electricity (ENTSO-E) for low-, medium-, and high-voltage electricity markets in Denmark. We derive production volume shares in all time steps for each market and treat these ENTSO-E samples as ground truth (grey). Then, we fit lognormal (red) and Dirichlet (green) distributions to the ENTSO-E data and visually compare their goodness-of-fit. Underlying data for this figure are available in Supporting Information S3.

2.6 | LCA case study

While the presented sampling modules can be used in any LCA study, the GSA results and LCIA scores are generated for carbon footprint of an average Swiss household consumption (Froemelt et al., 2018). The consumption foreground inventory model has ecoinvent v3.8 with the cutoff system model in the background, with 191'226 and 223'956 uncertain exchanges from technosphere and biosphere, respectively (Wernet et al., 2016). The climate change LCIA method with the time horizon of 100 years contains 194 greenhouse gases with global warming potential values that contain uncertainties as described in the 2013 IPCC report (Stocker, 2014). Python code and Brightway projects that implement the Swiss consumption model are available at Kim (2024b), and the code for adding uncertainties to climate change characterization factors can be found at Kim (2024a).

In this GSA study, we aimed to account for uncertainties across various components of the LCA model. Ecoinvent was selected because it is widely used and provides uncertainty distributions and parameterization of datasets. Climate change was chosen as an impact category because it is one of the few for which methodology and data are available to incorporate uncertainty into characterization factors (Stocker, 2014). The Swiss consumption model was selected due to its inclusion of a broad range of datasets, making it an interesting and challenging case for GSA given the large number and diverse nature of uncertain inputs. Additionally, building on our previous work with a similar model allows for a more direct comparison of results between independent and correlated sampling cases, leading to a deeper understanding of the outcomes (Kim et al., 2022). It is important to note, however, that the case study itself and the selected databases are not the primary focus of this analysis; the emphasis is on the methods developed. While the MC and GSA results will vary in other LCA studies, our open-source code is designed to facilitate its application to any database and case study that follows the Brightway structure.

3 | RESULTS

3.1 | Validation of Dirichlet distributions on electricity data

Figure 6 depicts Dirichlet samples for low-, medium-, and high-voltage markets in Denmark against electricity data from ENTSO-E that serves as ground truth, and lognormal distributions typically used in the ecoinvent database (see Section 2.3). The example of Danish markets is representative of other European countries, for which we provide similar plots in Section 4 of Supporting Information S1.

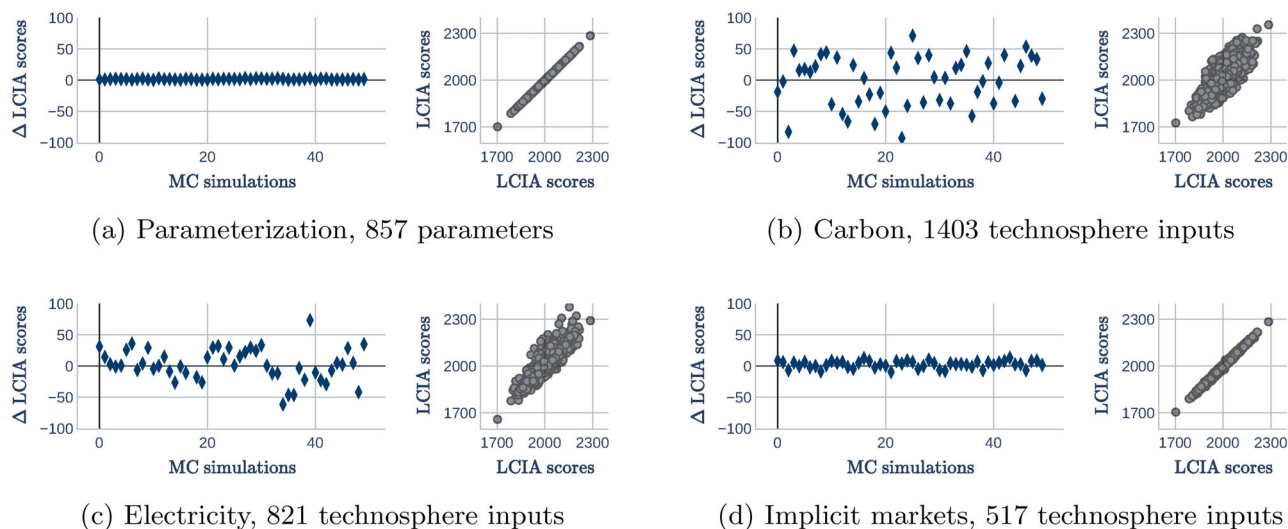


FIGURE 7 The effect of sampling modules on life cycle impact assessment (LCIA) scores for the carbon footprint of an average Swiss household. For each module, we show two subplots: (left) difference between LCIA scores obtained with independent and correlated sampling on the y-axis for the first 50 Monte Carlo (MC) simulations on the x-axis; (right) scatter plots of 2000 LCIA scores when inputs vary independently on the x-axis against scores when the respective sampling module is added on the y-axis. Underlying data for this figure is available in Supporting Information S3.

Our simulations show that Dirichlet samples for the low- and medium-voltage markets closely follow the reported data and offer a better representation of real-world conditions than the fitted lognormal distributions. For the high-voltage market, at first sight, lognormal distributions seem to fit the data quite well in the absence of multiple peaks and negative skewness (these can be seen in, e.g., “high voltage” market inputs). However, this is only possible because lognormals fit to each electricity supplier independently, which is a simpler task than satisfying constraints for multiple suppliers simultaneously. Dirichlet samples represent the real electricity data reasonably well for the suppliers with higher production volume shares, since the scaling factor λ was optimized for these exchanges (see Section 3 of Supporting Information S1). Due to the large number of suppliers in this market, it is impossible to select one λ to regulate the widths of all the marginal distributions as opposed to the cases of low- and medium-voltage markets. The multimodal nature of electricity measurement data is the reason why in reality fixed total sum is possible also in the presence of many suppliers. However, both lognormal and Dirichlet distributions fail to fully capture complex multimodal behavior that appears due to the intermittent nature of wind power dominating Danish electricity market, which also affects the dynamics of imports from other countries.

In general, the lognormals are not suitable for modeling uncertainty in markets for multiple reasons: (i) they cannot take zero values by definition, and yet it is common to have no electricity production; (ii) they are asymmetric and positively skewed, hence cannot portray well higher shares that are skewed negatively; (iii) they are not constrained to the $[0, 1]$ range, which renders it impossible to guarantee fixed totals. At the same time, the Dirichlet sampling for markets can represent real data in the absence of multiple peaks reasonably well. For multimodal marginal distributions, further disaggregation of electricity into different seasons and times of the day may eliminate the peaks.

While the Dirichlet approach may not provide a perfect solution for modeling uncertainty in markets, it performs at least as well as the lognormal distribution. Unlike the lognormal, the Dirichlet distribution enforces the “fixed total” constraint and captures correlations between variables, allowing for more realistic representation of uncertainties. This makes it a more flexible and appropriate choice for modeling dependent proportions.

3.2 | MC simulations with dependent and correlated sampling modules

Figure 7 shows how the sampling modules described in Section 2.2 affect LCIA results on the example of carbon footprint of an average Swiss household. The deterministic score for monthly consumption is 1870 kg CO₂-eq. Each of the four subplots showcases the results from running MC simulations with and without the respective sampling module, and independently of the other three modules. For each subfigure, high absolute differences (Δ LCIA score) between independent and correlated sampling on the left indicate higher influence of the respective sampling module. Only the scores from the first 50 MC simulations are plotted for the sake of visual analysis. For the subplots on the right, the scatter plot depicts LCIA scores when inputs vary independently on the x-axis against LCIA scores when inputs are correlated on the y-axis. The higher agreement is visually depicted as narrower spread of the data points.

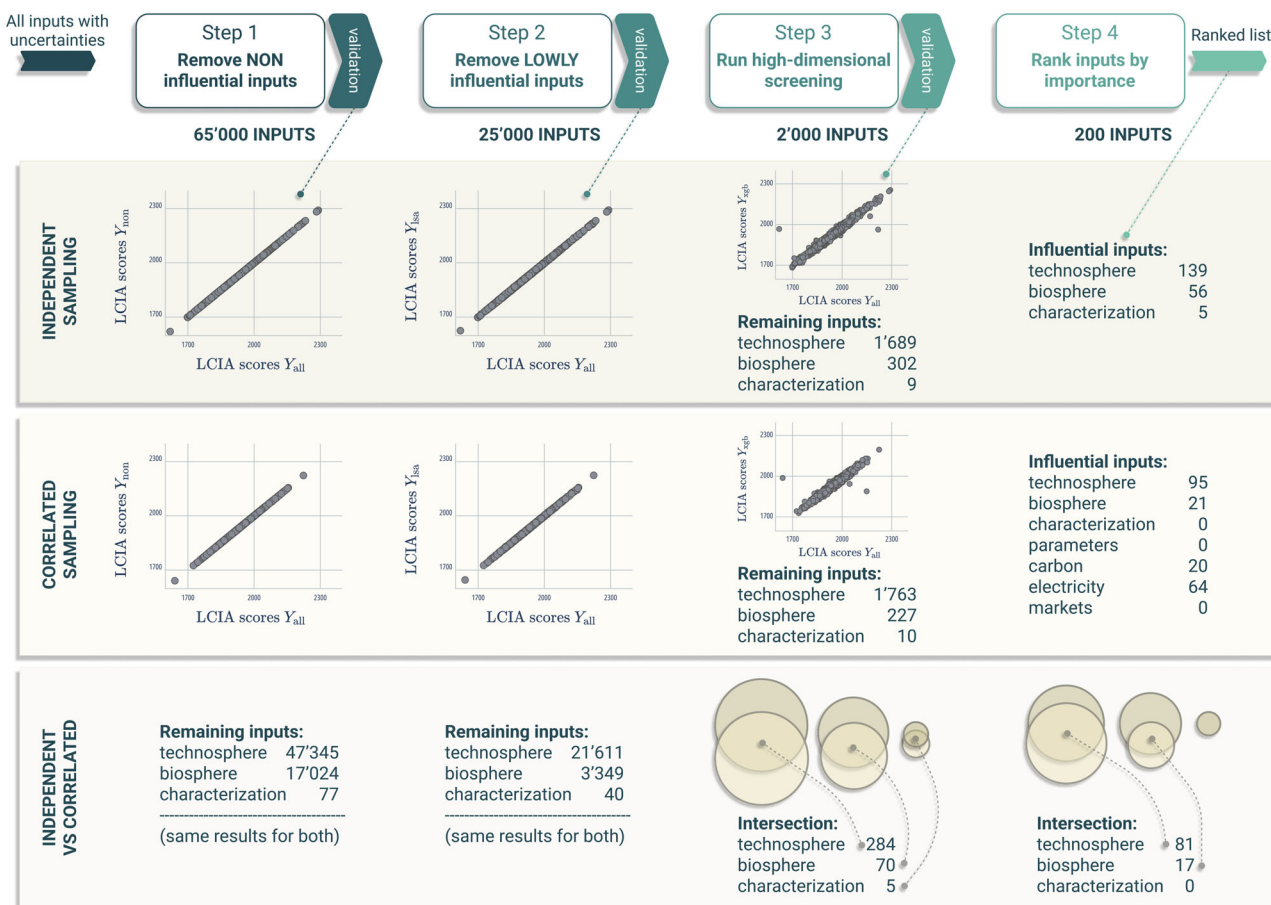


FIGURE 8 Summary of sensitivity results when Monte Carlo simulations are performed with independent and correlated sampling strategies. After each step, the figure provides the number of remaining potentially influential technosphere, biosphere, and characterization inputs and depicts life cycle impact assessment (LCIA) scores on validation plots as described in Section 2.5. Steps 1 and 2 are identical for both sampling cases, whereas for Steps 3 and 4, the intersection of the remaining inputs is given.

We observe the biggest impact on LCIA results by incorporating carbon balancing in liquid fuels and adding uncertainty to electricity markets, which is par for the course for a climate change study. On the contrary, the effect of parameterized ecoinvent inventories is negligible because many of the parameterized datasets are not relevant for the carbon footprint of Swiss consumption. These include land use activities and many biosphere flows that are anyhow irrelevant for the chosen impact category. Surprisingly, implicit markets do not significantly affect uncertainty in LCIA results, likely because many of them are not relevant for the given study or for the Swiss context (only 14 markets have location "Switzerland"). For a different LCA study using the same ecoinvent background database, the derived sampling modules can be used as is, though their effect and importance may vary. The reader is referred to Supporting Information S1 for the complete list of exchanges in each of the sampling modules.

3.3 | GSA results

We investigate the effect of adding causation and correlations on GSA by comparing results from independent and correlated sampling. Figure 8 summarizes sensitivity results for both cases and provides validation scatter plots after each step of the proposed GSA protocol (see Sections 2.4 and 2.5). The final GSA outcome of the top 200 influential inputs is given in Supporting Information S2.

After the first two steps, it was possible to filter out almost 95% of non- and lowly influential inputs, and validation plots confirm that no important inputs have been lost. While tuning XGBoost model in Step 3, we obtained $R^2 = 0.79$ for independent sampling, and $R^2 = 0.84$ for correlated sampling with the tuning parameters presented in Section 5 of Supporting Information S1. While the models can be tuned for higher performance, this is overall a good result for the Swiss household consumption because it incorporates many sectors with different underlying processes (food, housing, transport, durable goods, etc) (Froemelt et al., 2018). Feature importance in the boosting model allowed us to further reduce the number of potentially influential inputs to 2000, which were ranked on the subsequent step with the SHAP method.

There is a general agreement between sensitivity results based on two sampling approaches, with the overlap of 81 technosphere and 17 biosphere exchanges among the top 200 most influential inputs. Most of the LCIA scores' uncertainty is attributed to technosphere, and the important categories include: (i) electricity voltage transformation and heat production; (ii) passenger cars operation and petroleum production; (iii) computing and electronics manufacturing; (iv) agricultural processes, including milk, cheese, and meat production. Regarding biosphere, important are carbon dioxide emissions from heat and electricity production and from using passenger cars.

The main difference in ranking comes from the carbon balancing and electricity modules. For the former, 20 technosphere inputs were ranked high in the correlated sampling case. When using independent sampling, GSA yields 10 carbon dioxide emissions from "transport, passenger car" activities in Europe that are related to the liquid fuels combustion module. However, it fails to recognize that the underlying cause is the combustion of petrol and diesel, which was correctly identified in the correlated sampling case. Regarding the electricity module, the total of 60 technosphere inputs were ranked as top 200. GSA clearly indicates that uncertainty in electricity mixes matters, but mostly on the high voltage level and in the neighboring to Switzerland countries for the given LCA study. This includes imports between countries and relevant generation types within each country. Predictably, ecoinvent parameterization and implicit markets did not contain influential inputs, which was already hinted by the MC results (Section 3.2).

Interestingly, GSA with independent sampling identified as important five characterization factors and eight "carbon dioxide from soil or biomass stock" biosphere exchanges. These were proven to be important in Kim et al. (2021), where the characterization factors exhibit wide distributions and the carbon dioxide exchanges likely have an incorrectly defined uncertainty in the ecoinvent database. Yet, in the GSA with correlated sampling, these factors have dropped in importance, no longer appearing in the top 200 but not even among the first 2000 significant inputs. This shift in ranking highlights a critical insight: When correlations are properly accounted for, the overall uncertainty in LCIA scores is more heavily influenced by combustion and electricity inputs, whose contributions overshadow errors in the ecoinvent database. In practical terms, this suggests that addressing uncertainties in these primary sources—combustion and electricity—should take precedence over those in ecoinvent. Therefore, this analysis can help LCA practitioners prioritize data improvements where they have the greatest impact on reducing uncertainty. By doing so, it enhances the overall accuracy of the analysis and improves its reliability by minimizing the risk of over- or underestimating uncertainty. This is particularly important, as LCA is often used to inform decisions that involve substantial investments, making it crucial to ensure that results are accurately quantified.

4 | DISCUSSION

From each of the sampling modules, we could gain insights and observe limitations of typical inventory modeling:

- (i) Parameterization and carbon balancing revealed that the default ecoinvent uncertainty does not always match with the distribution obtained from varying the underlying independent inputs. We could not track any systematic relation between the two, and the width of the uncertainty from independent inputs was arbitrarily higher, lower, or well aligned with the predefined distribution. It is likely a consequence of using the pedigree approach based on expert judgement instead of fitting distributions to samples (Muller et al., 2016). This discrepancy should be further investigated to achieve higher quality uncertainty data. It is possible that the uncertainty comes not only from the defined independent inputs but also from other unspecified conditions. Then, the discrepancy can be corrected with additional error terms defined as separate inputs.
- (ii) The carbon balancing module had a high impact on LCIA scores and was the most straightforward to implement. If the balances in activity inputs and outputs are respected at the creation of inventories and in the deterministic LCIA, then we can use their ratios in the MC simulations and extend this sampling module to many more processes. Such sampling should become part of the standard uncertainty propagation.
- (iii) For the electricity mixes module, ENTSO-E does not provide as diverse electricity production types as ecoinvent, as shown in Section 1 of Supporting Information S1. Therefore, the disaggregation of the ENTSO-E data to ecoinvent categories is subject to choices and should be studied further. Moreover, Hirth et al. (2018) have reported on the shortcomings of the ENTSO-E Transparency Platform regarding data quality, which revealed gaps in electricity data, inconsistencies in data reporting, discrepancies when checked against other sources, and lack of documentation. Nonetheless, uncertainties in electricity markets can significantly impact GSA of LCA, highlighting the importance of including them in the analysis. We, therefore, advise using this sampling module with caution.
- (iv) Similarly to electricity, ENTSO-G is a Transparency Platform that issues regular information on gas supply and demand in Europe (ENTSO-G, 2022). Its gas data should also be used in combination with ecoinvent inventories, and they can be collected using Python client `ENTSO-G-PY`¹.
- (v) Dirichlet distributions from the implicit markets can be generally used whenever fractions should sum to totals. For lower number of variables and in the absence of multimodal marginal distributions, Dirichlet samples match real data quite well. For larger number of variables, the method should be further improved by fitting Dirichlet multiple times either to partial data or to subsets of variables.

Regarding the GSA protocol, we could decrease computation times by approximately a factor of 2, taking the procedure of Kim et al. (2022) as the basis and combining it with the efficiency of the technosphere inputs screening from Cucurachi et al. (2021). It was possible to run all computations on a personal computer within 24 h without cluster computing. The computational demand for the correlated sampling case is the same as for the independent sampling, but the implementation of new types of sampling modules might not be straightforward. Gradient boosting gave satisfactory results already with the default tuning parameters for LCA models (Kim et al., 2021), which was a good start for the finer grid search parameter tuning. Notably, to the extent of our knowledge, the SHAP method has not been widely adopted in GSA of LCA, even though it is natural for the investigation of inputs importance in the presence of complex nonlinear relationships.

5 | CONCLUSION

Despite the diverse nature of dependent and correlated sampling, it is feasible to implement it. In this work, we showed it by developing modules with various relations, such as: (i) between parameters and technosphere and/or biosphere flows, (ii) from technosphere to biosphere, (iii) correlations that exist in measurements, and (iv) datasets with joint distributions. They serve as a starting point for further advancements in MC sampling for LCA models. Although we have presented only few types of correlated and dependent sampling, it is apparent that neglecting uncertainties and correlations can result in a different GSA outcome. Practically, it can lead to misleading conclusions about the relative contribution of different processes to total environmental impacts, potentially distorting the interpretation of LCIA results. Without properly accounting for correlations in modeling, the LCA may fail to accurately reflect uncertainty and the key sources of impacts, leading to misplaced efforts on data improvements and reducing the reliability of decision-making based on such analysis. Given that implemented modules are independent from the chosen LCA case study and LCIA method, they can be included into uncertainty and sensitivity analysis of more LCAs without substantial additional effort. The clear need in developing more advanced sampling approaches can be addressed with the BRIGHTWAY version 2.5 Python library, which allows to seamlessly include new sampling strategies atop the already existing ones (Mutel, 2022). The important question of “Which modules should be prioritized?” can be answered by GSA itself, as it can point to the bottlenecks in current inventory modeling. The wide diversity of possible correlated and dependent structures in state-of-the-art LCAs also poses a question, whether paradigm shift in inventory and impact assessment modeling is necessary to allow for the more native integration of executable models, multivariate distributions, and interfaces to stream measurement data as a standard practice in life cycle data collection. Although the methods developed in this study aim at improving the robustness of LCA models, the methodological advances can be integrated with data distribution, imputation, and uncertainty management in the broader field of industrial ecology.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the Ecoinvent Association. Restrictions apply to the availability of these data, which were used under license for this study. Data are available at <https://ecoinvent.org> with the permission of the Ecoinvent Association or directly from the third party.

The aggregated energy data of the control block Switzerland are publicly available at <https://www.swissgrid.ch/de/home/customers/topics/energy-data-ch.html>. The data on the electricity generation, transportation, and consumption for the pan European market are published on the Transparency Platform of the European Network of Transmission System Operators for Electricity (ENTSO-E): <https://transparency.entsoe.eu/>.

The Swiss household consumption model was derived from the work of Andreas Froemelt, David J. Dürrenmatt, and Stefanie Hellweg: <https://doi.org/10.1021/acs.est.8b01452>. The consumption model can be reproduced with the GitHub repository CONSUMPTION_MODEL_CH available at https://github.com/aleksandra-kim/consumption_model_ch.

The intermediate outputs of this work cannot be shared due to the license restriction of the Ecoinvent Association but can be recreated with the GitHub repository AKULA available at <https://github.com/aleksandra-kim/akula>. The final results of the conducted sensitivity analysis are available in the supporting information of this research article.

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ENDNOTE

¹<https://github.com/nhcb/entsog-py>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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